

The University of Azad Jammu and Kashmir, Muzaffarabad

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Roll NO: 2024-SE-06

Semester:4th

Course Title: Database System

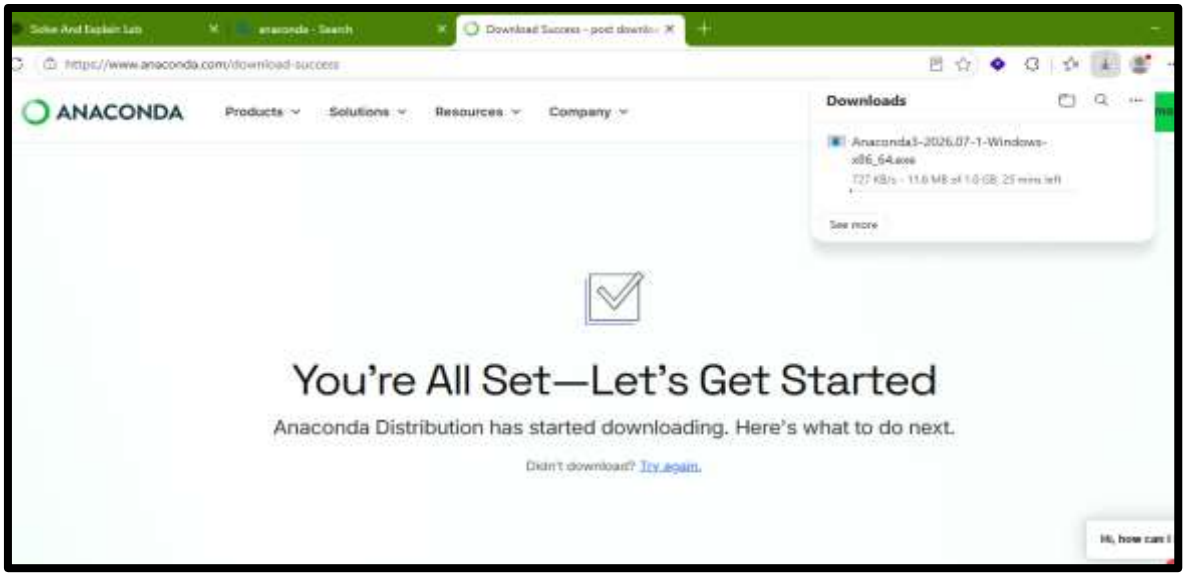
Course Code: CS-2204

Submitted To: Engr. Awais Rathore

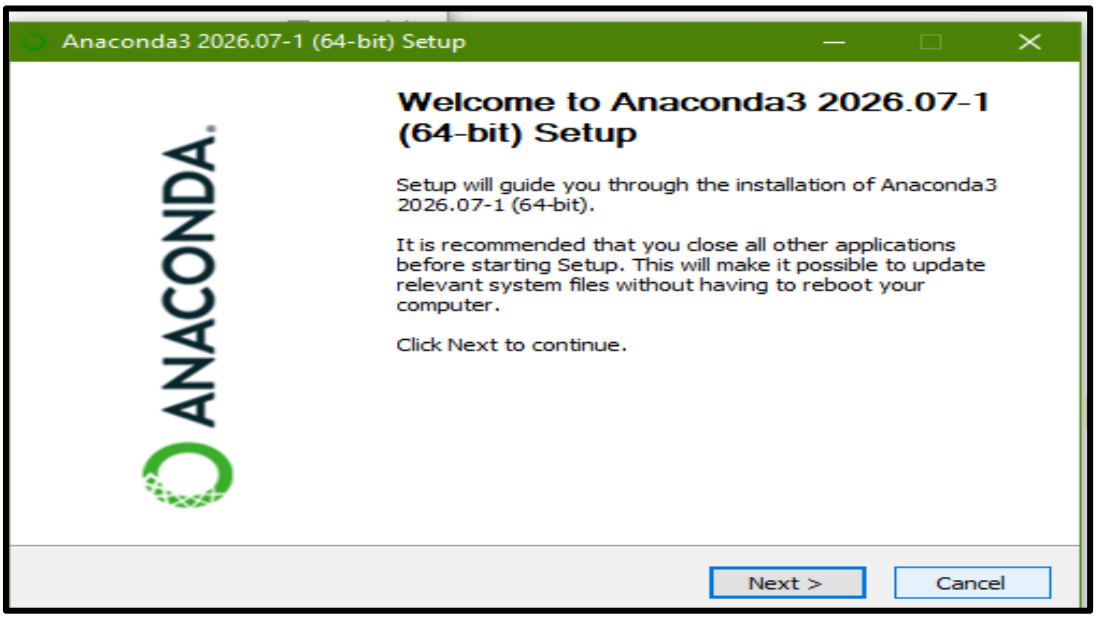
Department of Software Engineering

➢ Task 01: Installation Of Anaconda

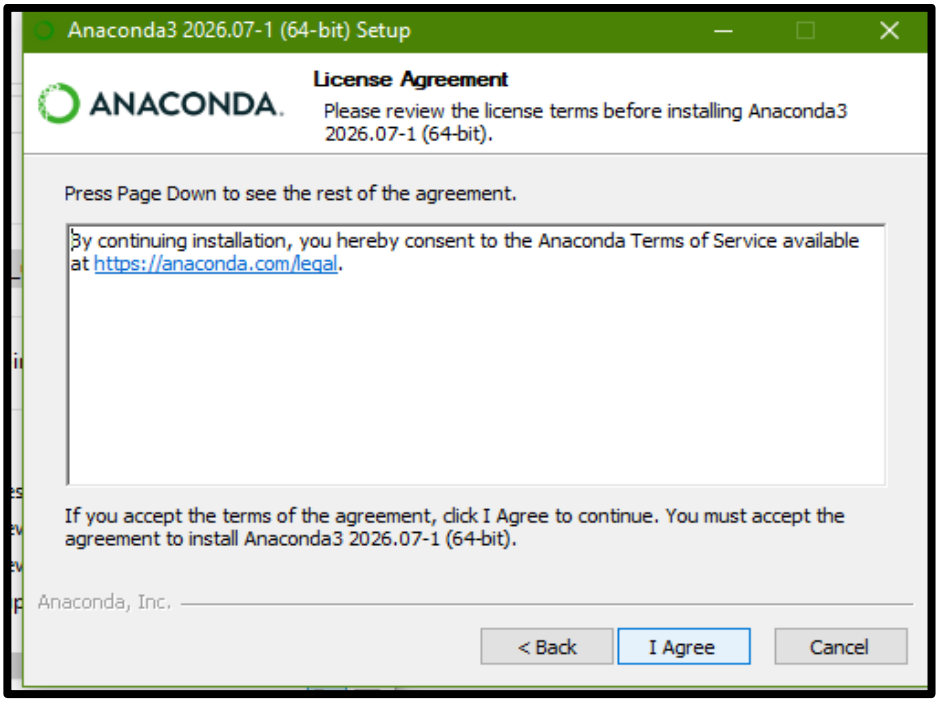
- **Step:01**



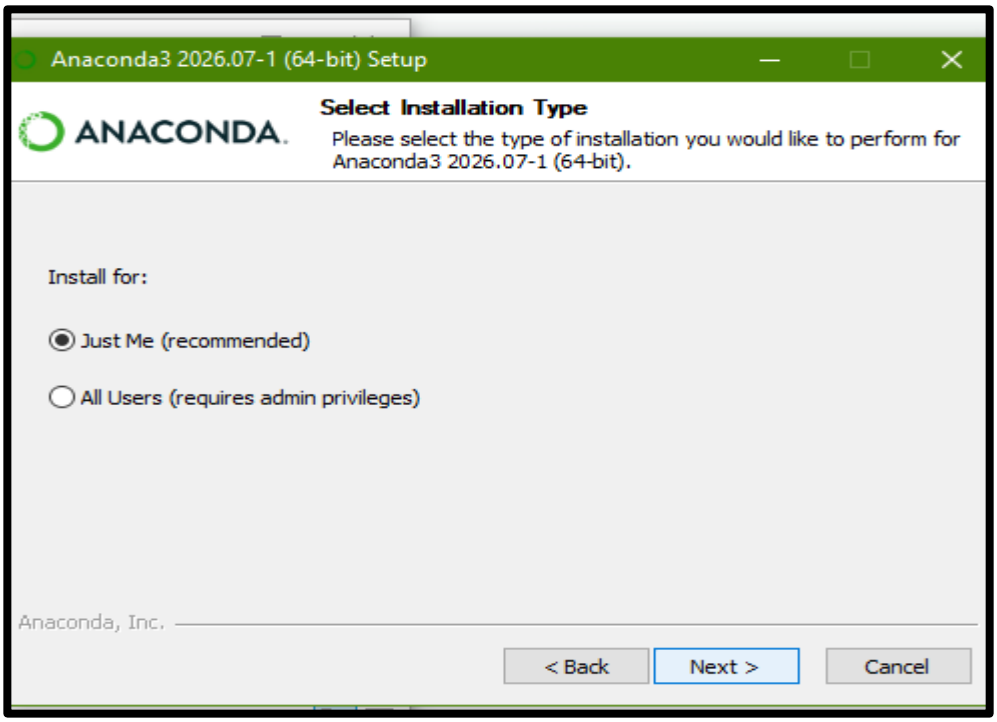
- **Step: 02**



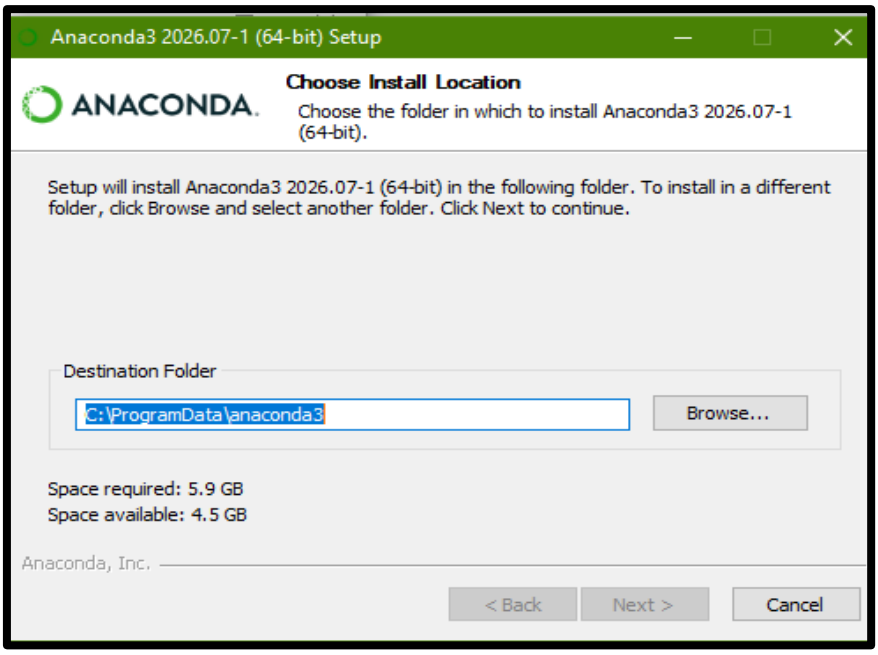
- **Step: 03**



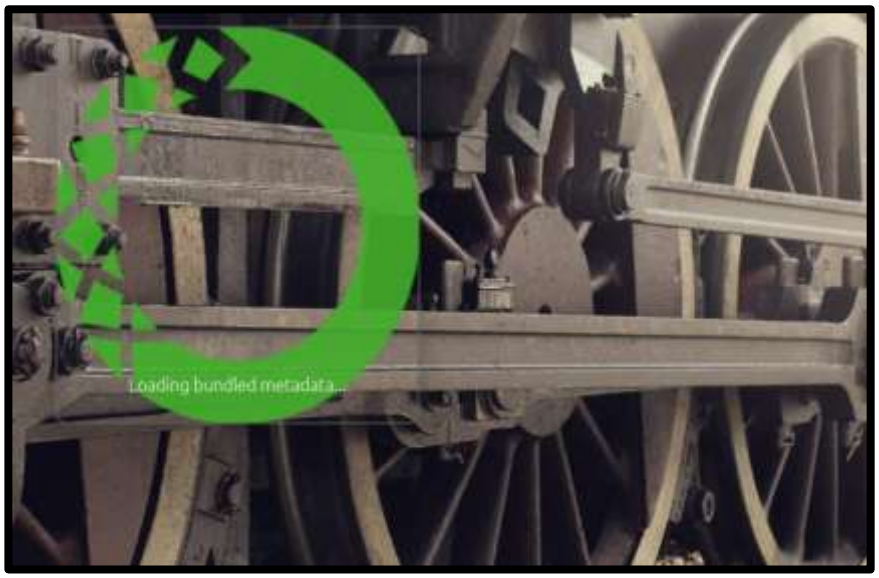
- **Step: 04**



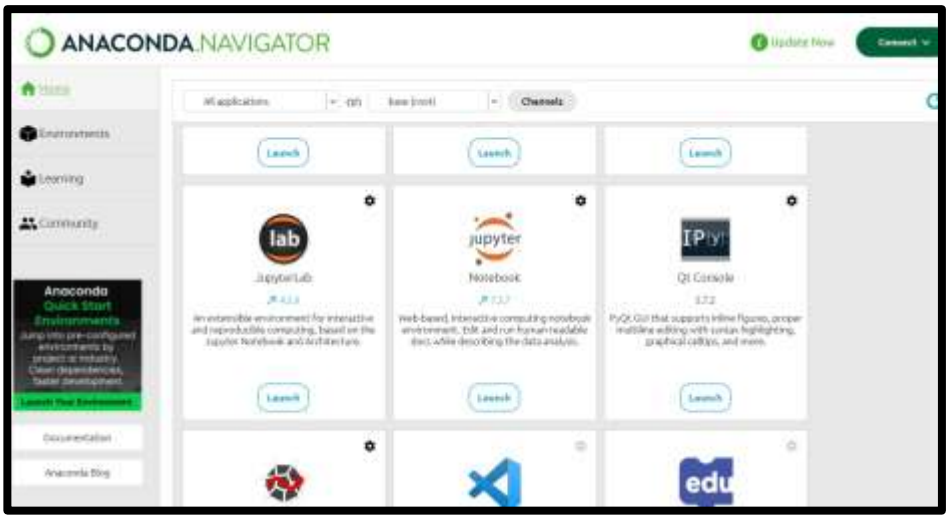
- **Step: 05**



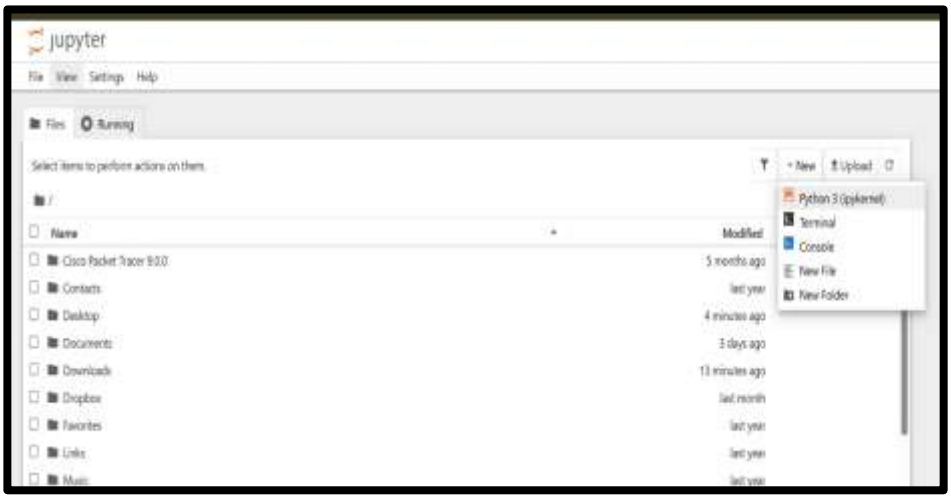
- **Step: 06**



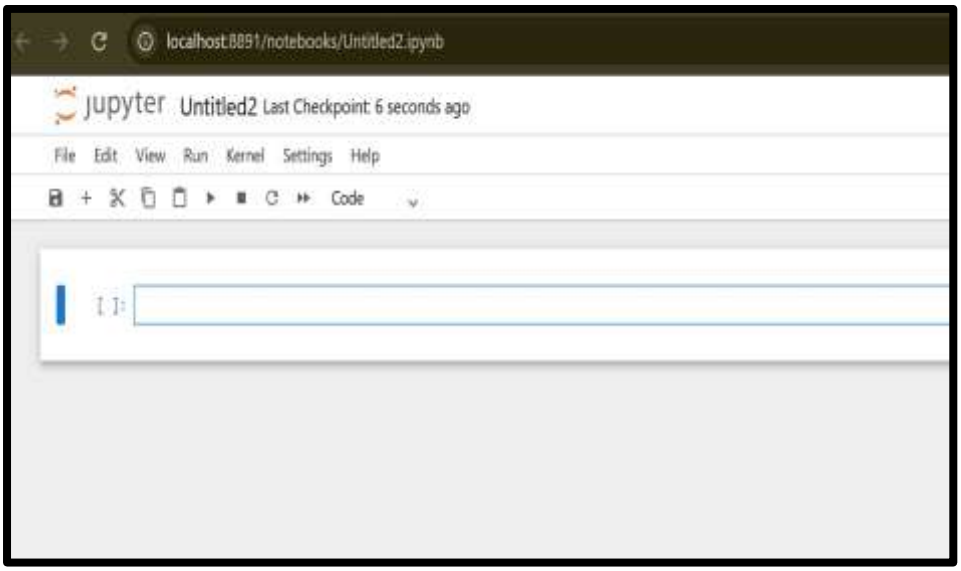
- **Step: 07**



- **Step: 08**



- **Step: 09**



➢ Task 2: Create three Python files

File name	Purpose
hello.py	Run your first Python program
basics.py	Solve all six exercises
lab01_imports.py	Import the required libraries

➢ File 01: hello.py



➢ File 02: basics.py

➢ Exercise 1 — Variables and data types

Create variables for image width, height, name, and color status. Print their values and types, then calculate the total pixels.

Exercise 01: Variables And Datatypes

[7]:

```
width = 640
height = 480
image_name = "sample.png"
is_color = True

print(width, type(width))
print(height, type(height))
print(image_name, type(image_name))
print(is_color, type(is_color))

total_pixels = width * height
print("Total pixels:", total_pixels)
```

```
640 <class 'int'>
480 <class 'int'>
sample.png <class 'str'>
True <class 'bool'>
Total pixels: 307200
```

➢ Exercise 2 — if, elif, and else

Label an intensity as Invalid, Bright pixel, or Dark pixel.

An intensity below 0 or above 255 is invalid. A valid intensity of 128 or more is bright; otherwise, it is dark.

Exercise 02: if, elif, else

[9]:

```
intensities = [-1, 0, 127, 128, 255, 256]

for intensity in intensities:
    if intensity < 0 or intensity > 255:
        label = "Invalid"
    elif intensity >= 128:
        label = "Bright pixel"
    else:
        label = "Dark pixel"

    print(intensity, ":", label)
```

```
-1 : Invalid
0 : Dark pixel
127 : Dark pixel
128 : Bright pixel
255 : Bright pixel
256 : Invalid
```

➢ Exercise 3 — for and while loops

Use a for loop to label pixel values [20, 130, 255], then use a while loop to print 0, 1, and 2.

Exercise 03: For Loop

[11]:

```
pixels = [20, 130, 255]

for pixel in pixels:
    if pixel >= 128:
        print(pixel, "Bright pixel")
    else:
        print(pixel, "Dark pixel")
```

```
20 Dark pixel
130 Bright pixel
255 Bright pixel
```

Exercise 03: While Loop

[13]:

```
count = 0

while count < 3:
    print(count)
    count += 1
```

```
0
1
2
```

➢ Exercise 4 — Functions

Create a function called is_bright() that returns True if the pixel intensity is at least the threshold and False otherwise.

Exercise: 04

[14]:

```
def is_bright(pixel, threshold=128):
    return pixel >= threshold

print(is_bright(127))
print(is_bright(128))
print(is_bright(200, 210))
```

```
False
True
False
```

➢ Exercise 5 — Lists and tuples

Modify a list, print its length, and unpack a tuple into width and height.

Exercise 05: Lists And Tuples

```
[15]:
pixels = [0, 128, 255]

print("First item:", pixels[0])

pixels[1] = 150
pixels.append(64)

print("Updated list:", pixels)
print("Length:", len(pixels))

size = (640, 480)
width, height = size

print("Width:", width)
print("Height:", height)

# A list is mutable, so its items can be changed.
# A tuple is immutable, so its items cannot be reassigned.
```

```
First item: 0
Updated list: [0, 150, 255, 64]
Length: 4
Width: 640
Height: 480
```

➢ Exercise 6 — Dictionaries and sets

Add a mode to an image dictionary, access its values using keys, and count unique labels with a set.

Exercise 06: Dictionaries And Sets

```
[17]:
image = {"width": 640, "height": 480}
image["mode"] = "RGB"

print("Width:", image["width"])
print("Mode:", image["mode"])

labels = ["cat", "dog", "cat", "bird"]
unique_labels = set(labels)

print("Unique labels:", unique_labels)
print("Number of unique labels:", len(unique_labels))
```

```
Width: 640
Mode: RGB
Unique labels: {'dog', 'bird', 'cat'}
Number of unique labels: 3
```

➢ Installation Of Required Packages

The required libraries were confirmed using the Command Prompt command shown below.

```
python -m pip show numpy pillow matplotlib
```

Installed package details

```
C:\Users\Admin>python -m pip show numpy pillow matplotlib
Name: numpy
Version: 2.4.6
Summary: Fundamental package for array computing in Python
Home-page: https://numpy.org
Author: Travis E. Oliphant et al.
Author-email:
License-Expression: BSD-3-Clause AND 0BSD AND MIT AND Zlib AND CC0-1.0
Location: D:\anaconda\Lib\site-packages
Requires:
Required-by: astropy, bokeh, contourpy, datashader, h5py, holoviews, hvplot, ImageIO, imbalanced-learn, ipympl,
sy, pyerfa, PyWavelets, scikit-image, scikit-learn, scipy, seaborn, statsmodels, streamlit, tables, tifffile, x
```

Figure 13a. NumPy 2.4.6 - name, version and installation location.

```
Name: pillow
Version: 12.3.0
Summary: Python Imaging Library (fork)
Home-page: https://python-pillow.github.io
Author:
Author-email: Jeffrey 'Alex' Clark <aclark@aclark.net>
License-Expression: MIT-CMU
Location: D:\anaconda\Lib\site-packages
Requires:
Required-by: bokeh, ImageIO, ipympl, matplotlib, scikit-image, streamlit
---
```

Figure 13b. Pillow 12.3.0 - name, version and installation location.

```
---
Name: matplotlib
Version: 3.11.0
Summary: Python plotting package
Home-page: https://matplotlib.org
Author: John D. Hunter, Michael Droettboom
Author-email: Unknown <matplotlib-users@python.org>
```

Figure 13c. Matplotlib 3.11.0 - name and version.

➢ File 03: lab01_imports.py

Import verification

```
[3]: import sys

     print("Python version:", sys.version)
     print("Python executable:", sys.executable)

     Python version: 3.14.6 | packaged by Anaconda, Inc. | (main, Jul  9 2026, 14:29:05) [MSC v.1942 64 bit (AMD64)]
     Python executable: D:\Anaconda3\python.exe

[4]: import numpy as np
     from PIL import Image
     import matplotlib.pyplot as plt

     print("All three packages imported successfully!")

     All three packages imported successfully!

[ ]:
```